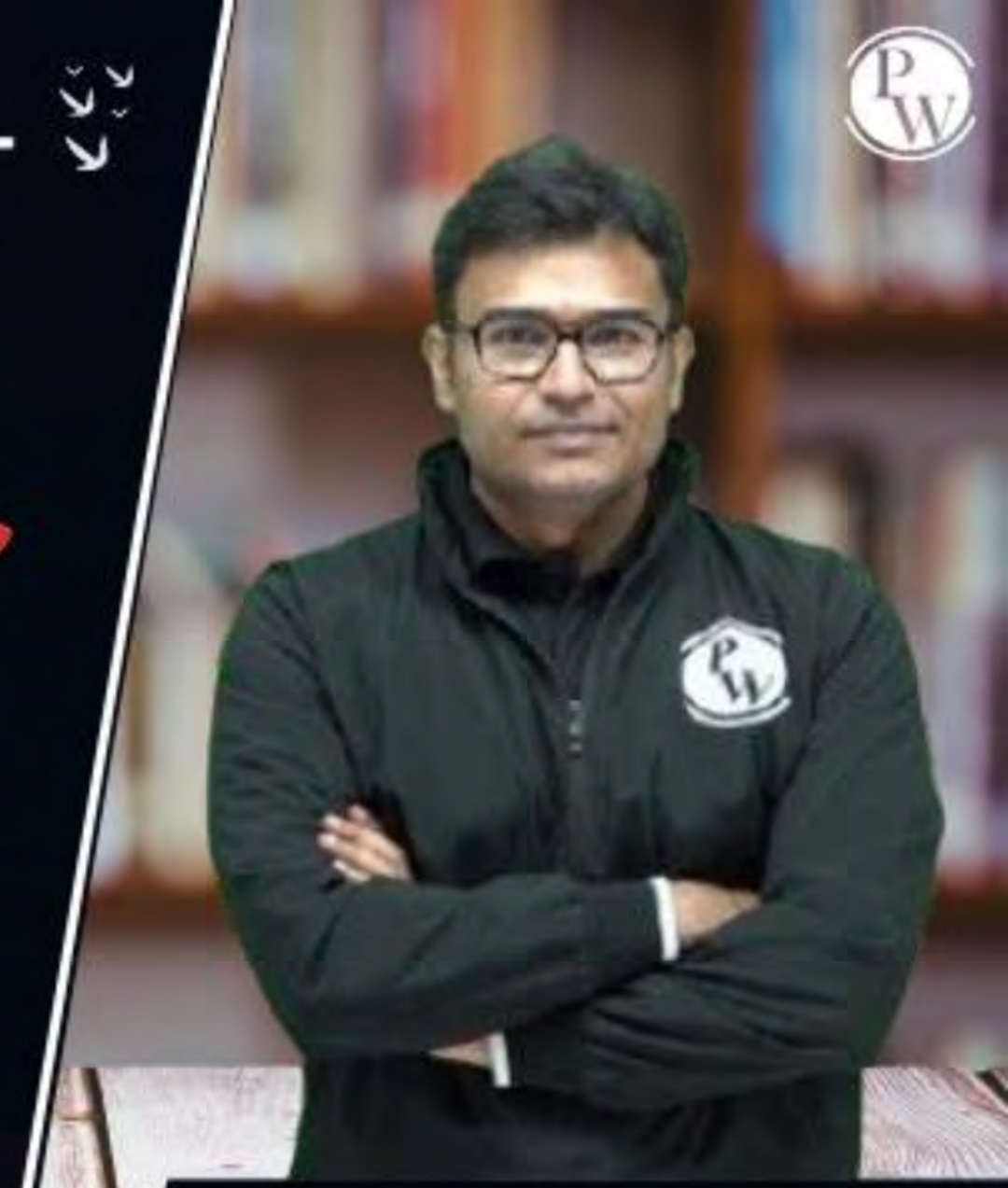
Computer Science & IT

Theory of Computation



Regular Languages

Lecture No. 04



By- Ravindra Sir

Recap of Previous Lecture



Topic

Topic

Topic

NFA DFA

Topics to be Covered



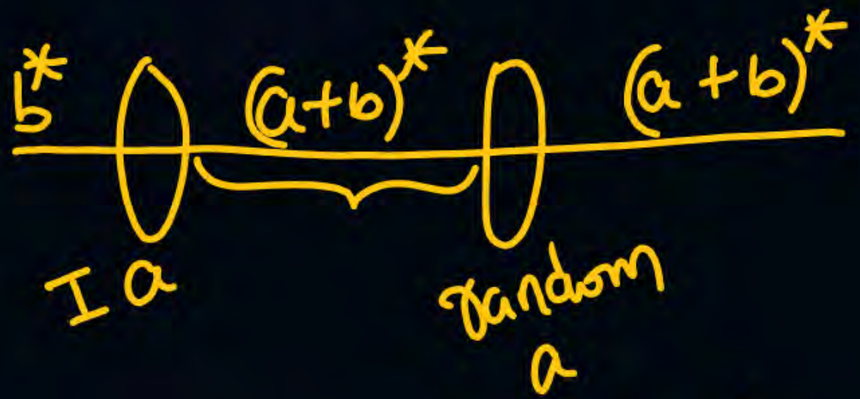
- Topic
- Topic
- Topic

DFA NFA

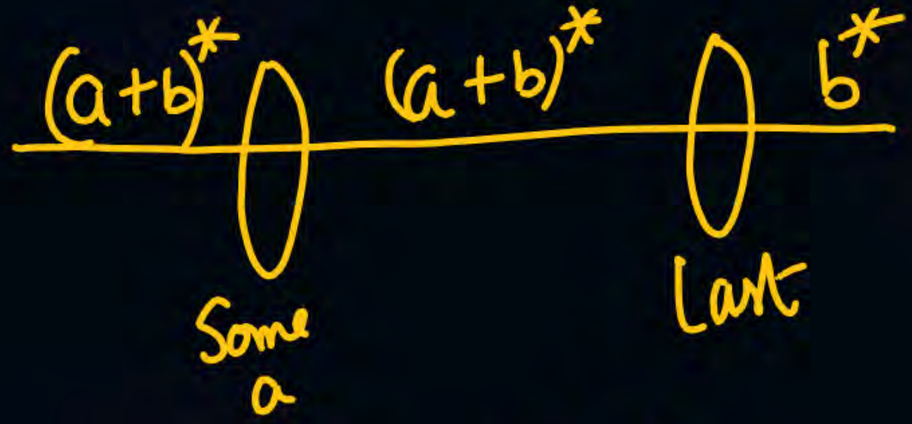
$\Sigma = \{a, b\}$ $L = \{ w / n_a(w) \geq 2 \}$



$(a+b)^* a (a+b)^* a (a+b)^*$



$$b^*a(a+b)^*a(a+b)^*$$



$$(a+b)^* a\,(a+b)^* a\,b^*$$

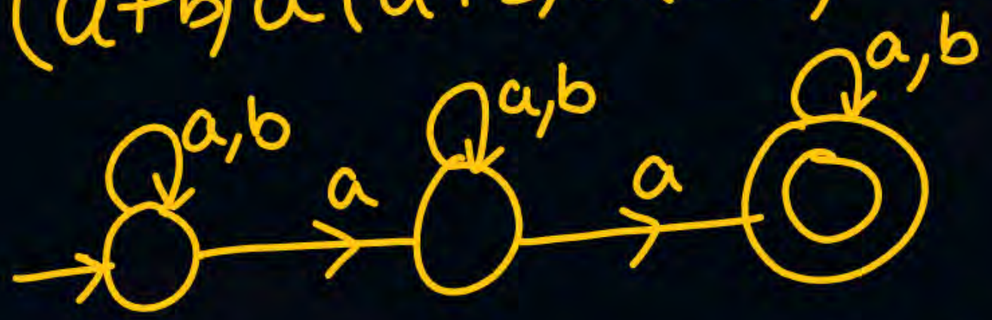
$$\underbrace{\quad}_{b^*} \; a \; \underbrace{\quad}_{b^*} \; a \; \underbrace{\quad}_{b^*}$$

$$(a+b)^* \quad (a+b)^* \quad (a+b)^*$$

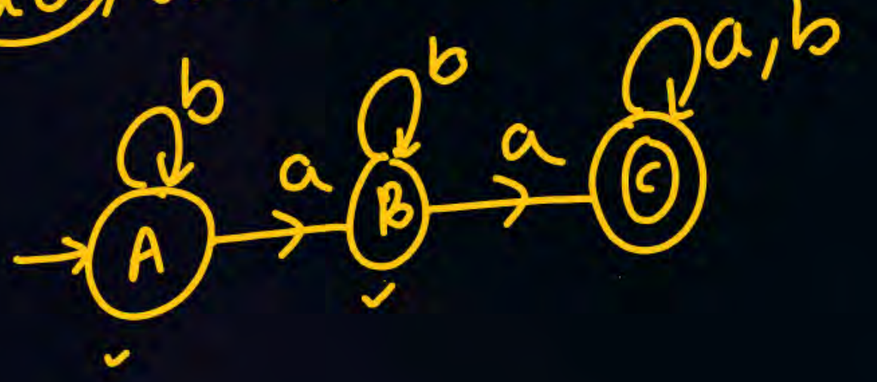
$$2 \times 2 \times 2 = 8 - 1 = 7$$

$(b^* a b^* a b^*) \Rightarrow$ exactly 2 a's

$(a+b)^*a(a+b)^*a(a+b)^*$



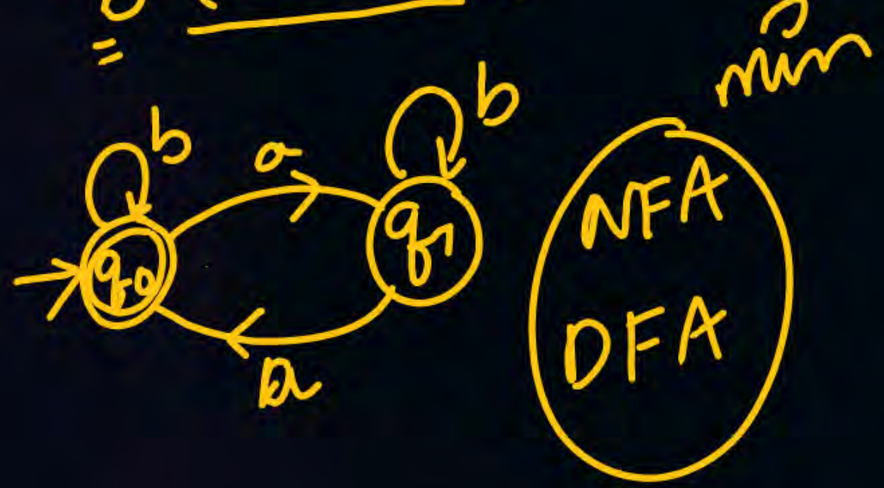
$L = \{aa, aab, baa, aba, aaa \ldots\}$



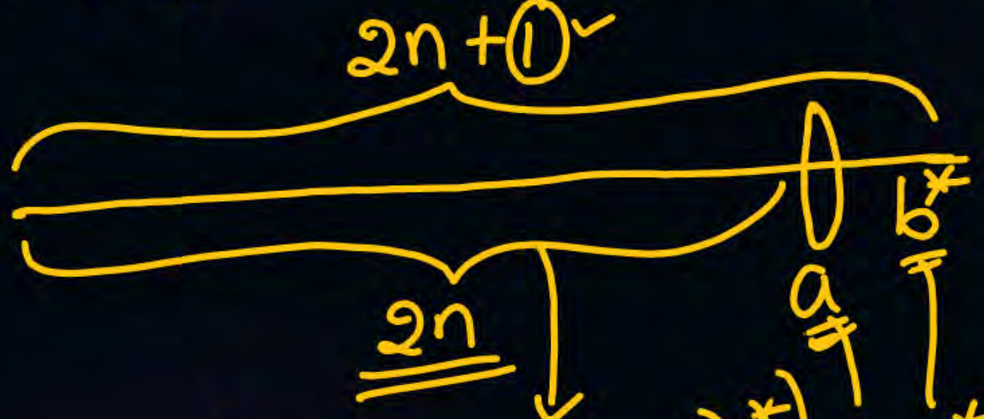
$\Sigma = \{a, b\}$ $L = \{\omega / n_a(\omega) \text{ are even}\}$

$b^* + (b^* a b^* a b^*)^*$

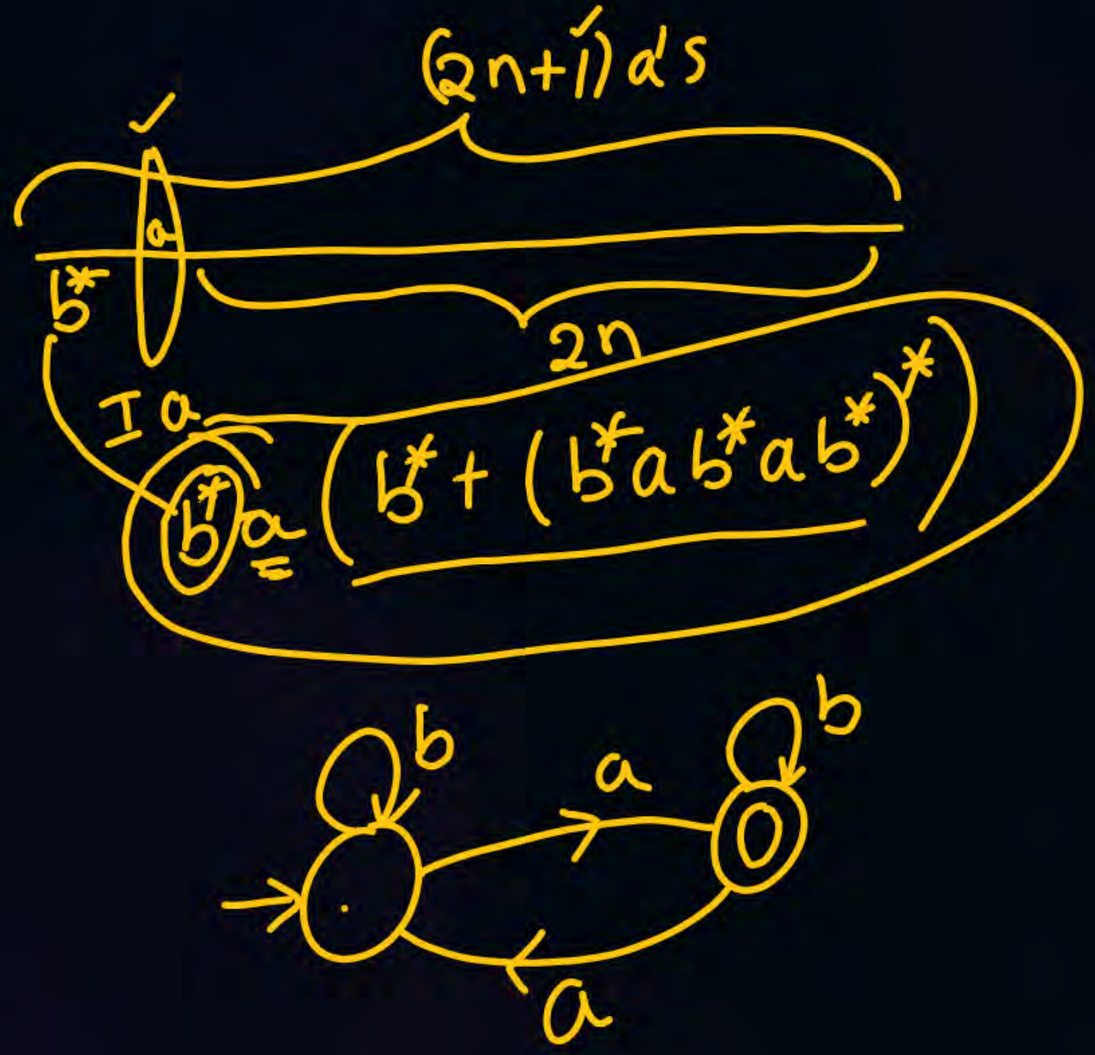
$b^* (a b^* a)^* b^*$

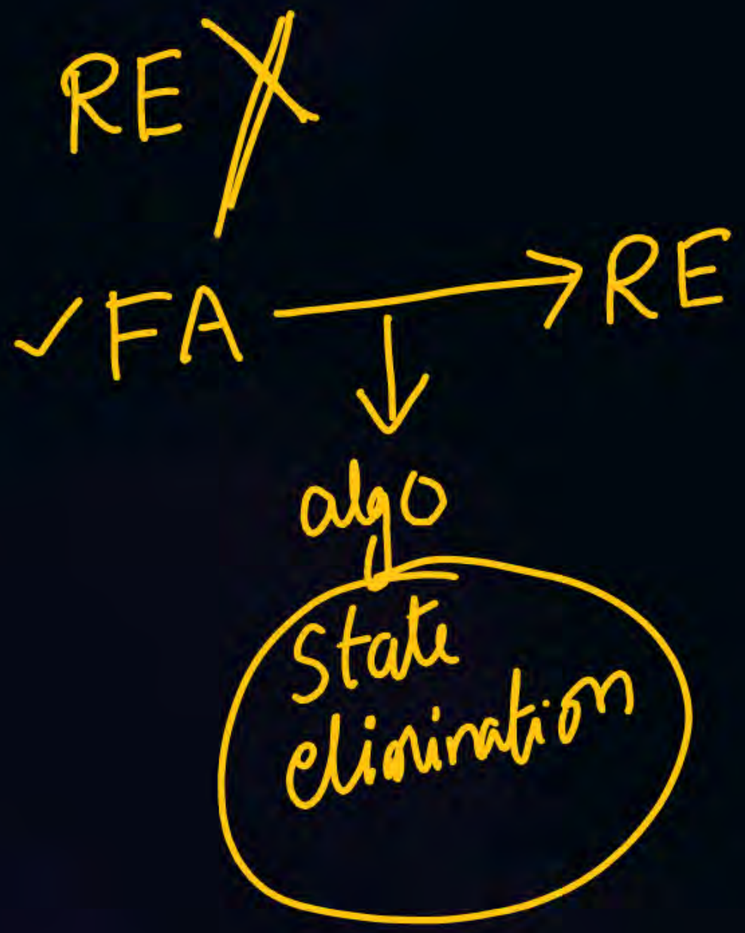


$\Sigma = \{a, b\}$ $L = \{w / n_a(w) \text{ are odd}\}$



$$(b^* + (b^* a b^* a b^*)^*) a b^*$$


(2n+1) a's
a
b*
2n
a
b*
a
(b* + (b*ab*ab*)*)
b
a
b
a


RE
FA
RE
algo
State elimination

$\Sigma = \{a, b\}$ $L = \{ w / n_a(w) \cong 0 \bmod 3 \}$

$b^* + (b^* a b^* a b^* a b^*)^*$



0, 3, 6, ⋮, 1

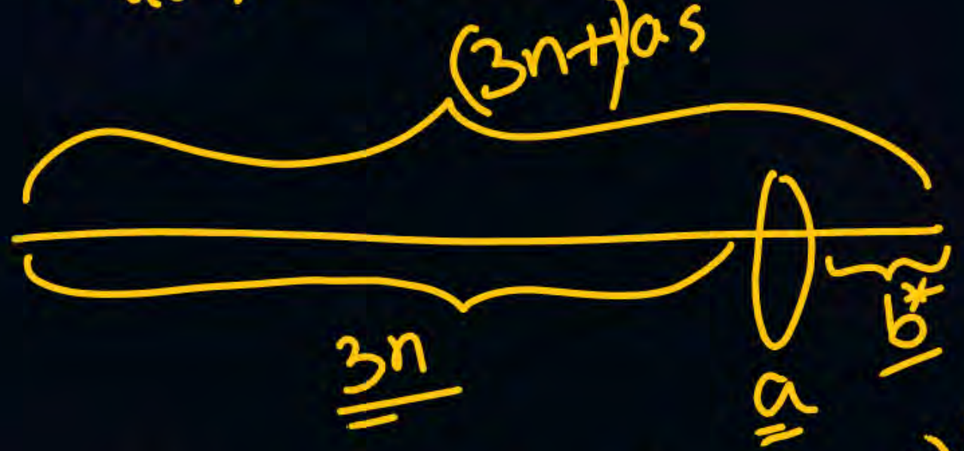
1, 4, 7, ⋮

2, 5, 8

$n_a(w) \cong 1 \bmod 3 \rightarrow q_1$

$n_a(w) \cong 2 \bmod 3 \rightarrow q_2$

$n_a(w) \equiv 1 \bmod 3$



$(b^* + (b^* a\, b^* a\, b^* a b^*)^*)\, a\, b^*$



$b^* a\, (b^* + (b^* a\, b^* a\, b^* a\, b^*)^*)$

$\cong 2 \bmod$

$$(b^* + (b^*ab^*ab^*ab^*)^*)ab^*ab^*$$

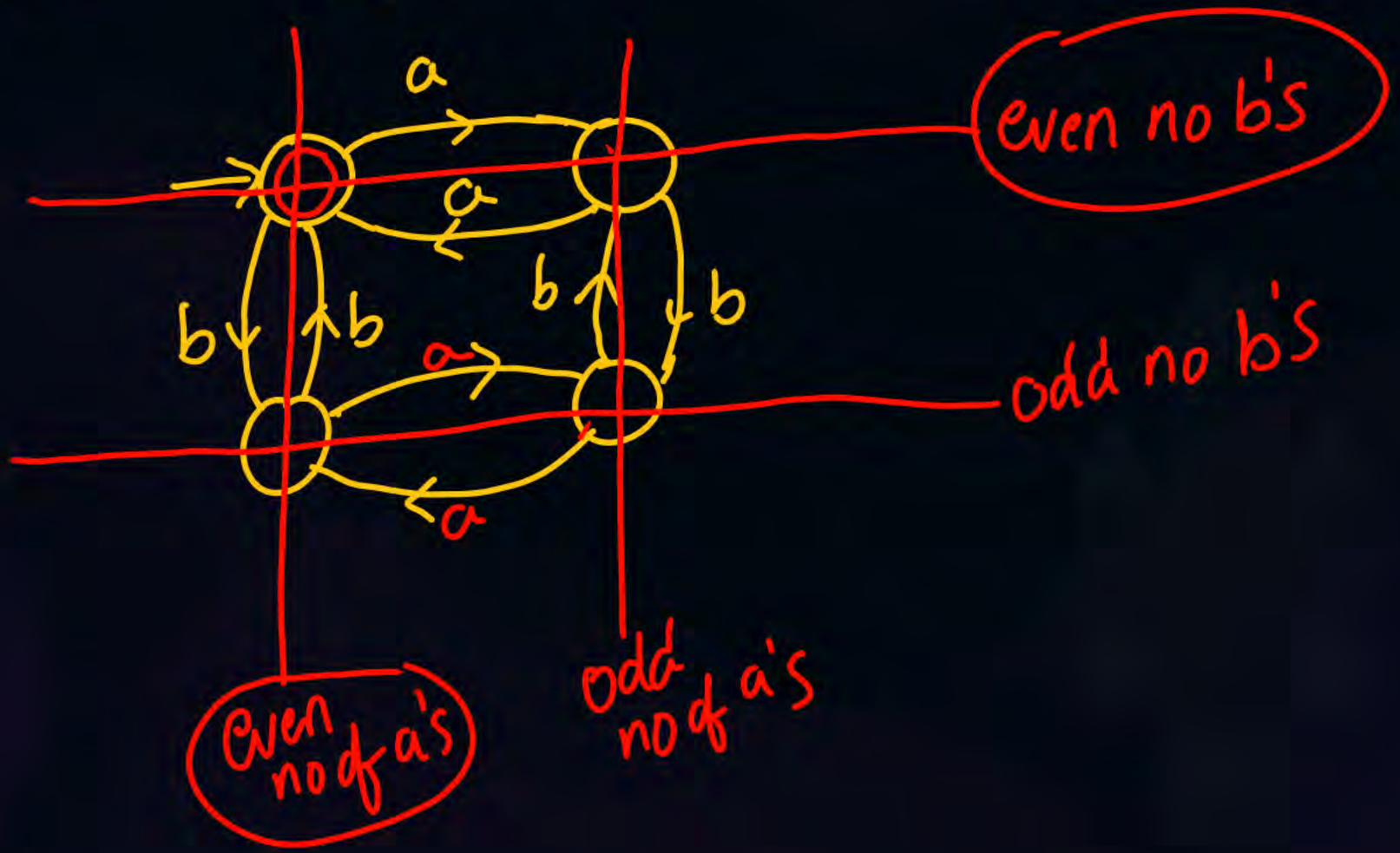
$$b^*ab^*a(b^* + (b^*ab^*ab^*ab^*)^*)$$

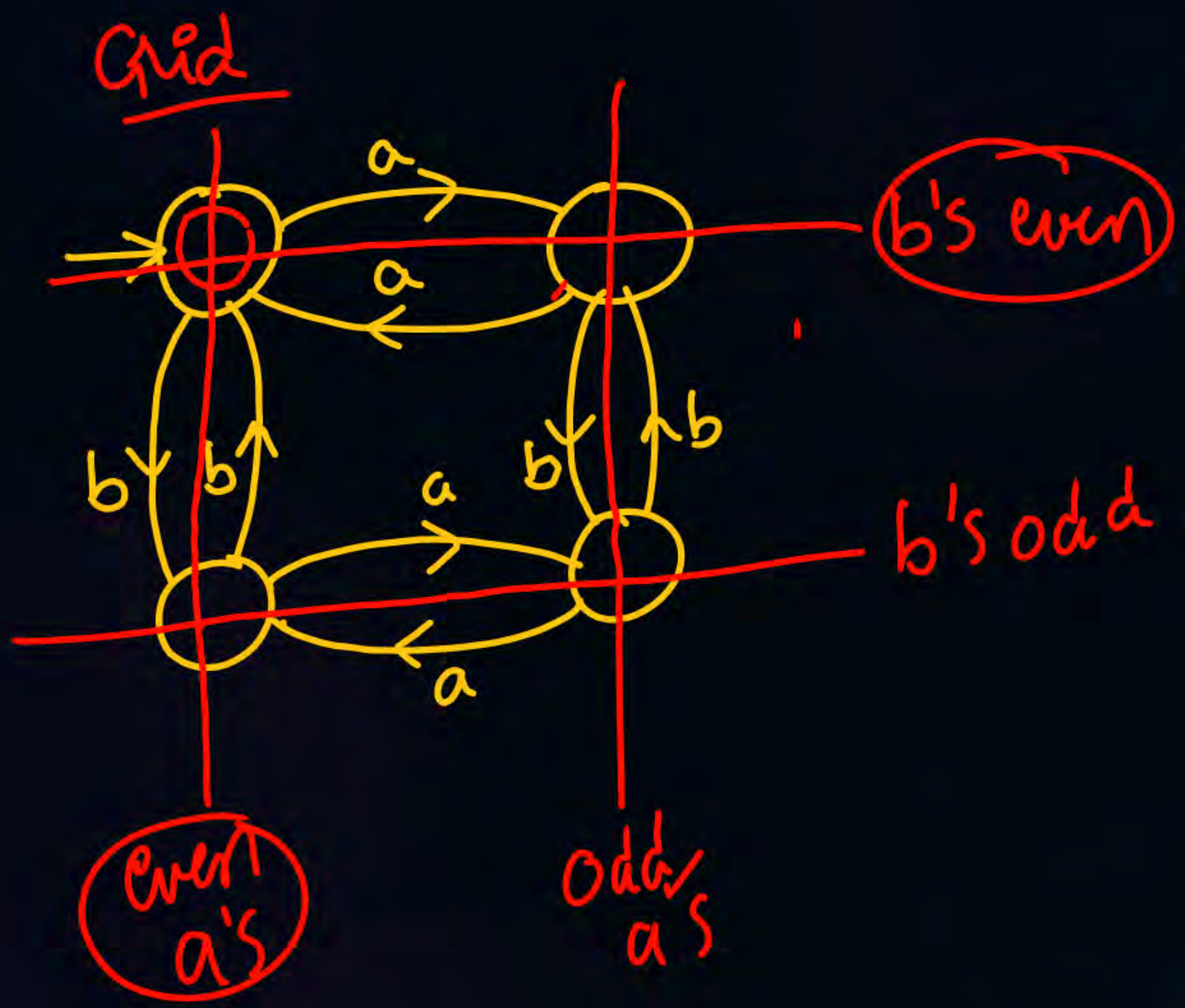
$$b^*a(b^* + (b^*ab^*ab^*ab^*)^*)ab^*$$

$\Sigma = \{a, b\}$ $L = \{w / n_a(w), n_b(w)$ are even$\}$

$\{E, O\} \times \{E, O\}$ 2 minutes

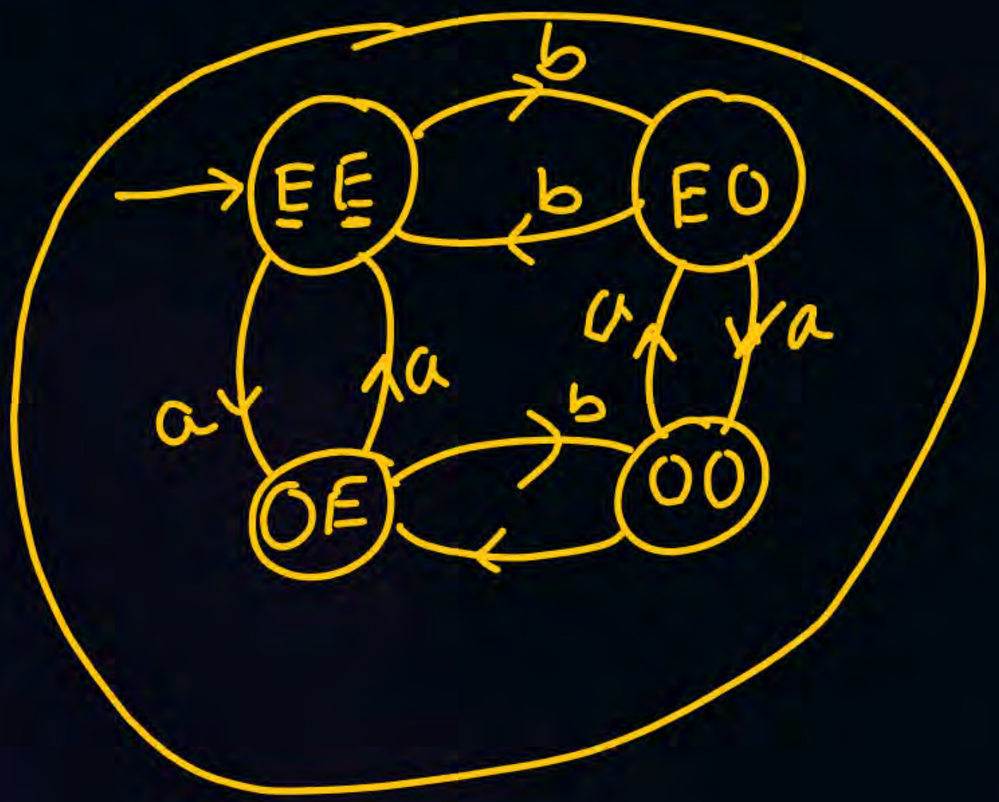
4



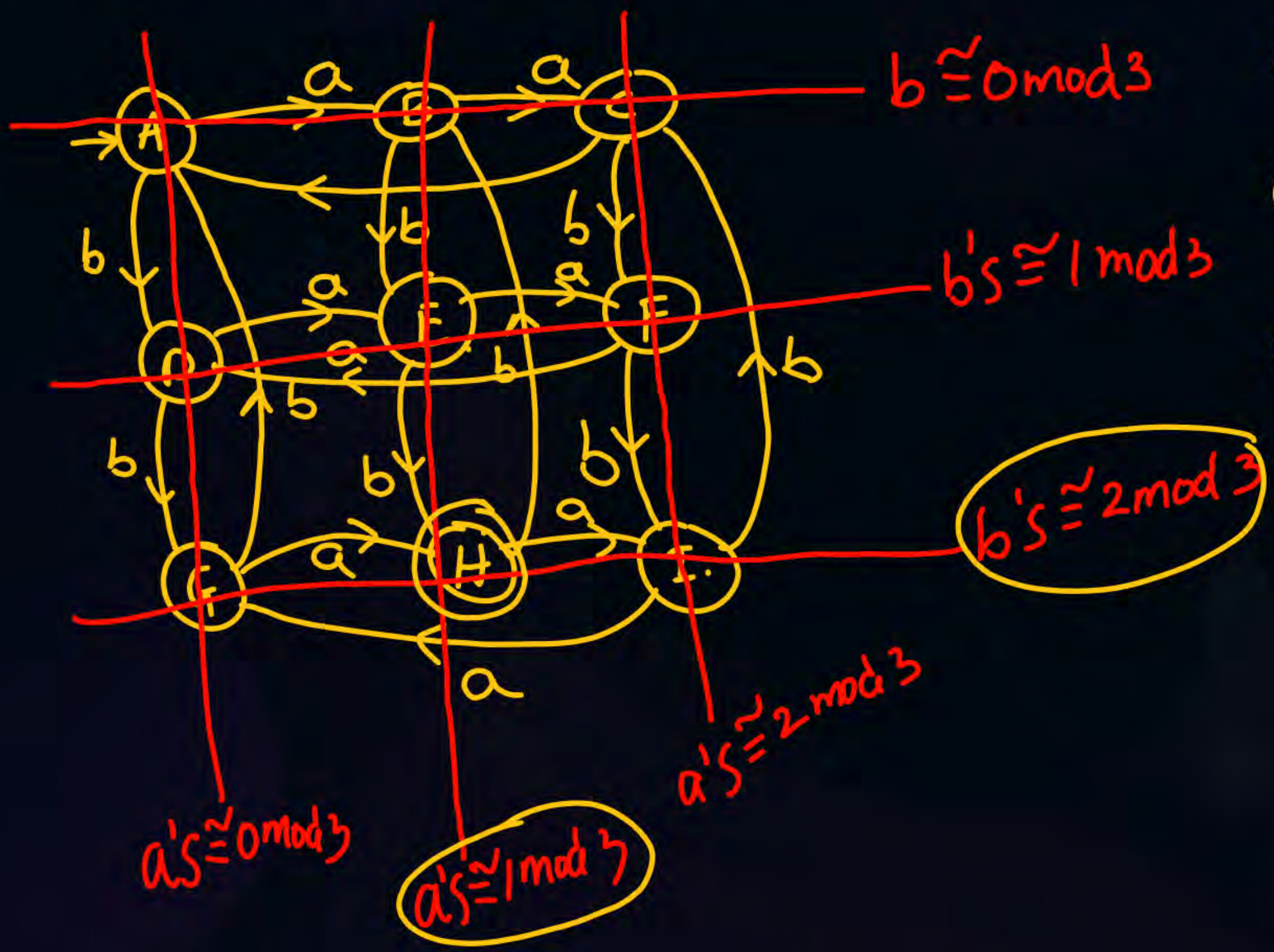

Grid
a
a
b's even
b
b
b
b
a
b's odd
a
even a's
odd a's

a's b's

$\{E, O\} \times \{E, O\}$ $\in$



$\Sigma = \{a, b\}$ $L = \{w / n_a(w) \cong 0 \bmod 3,\ n_b(w) \cong 0 \bmod 3\}$



$a \cong 0$ and $b's \cong 0$

A

$a's \cong 1$ and $b's \cong 1$

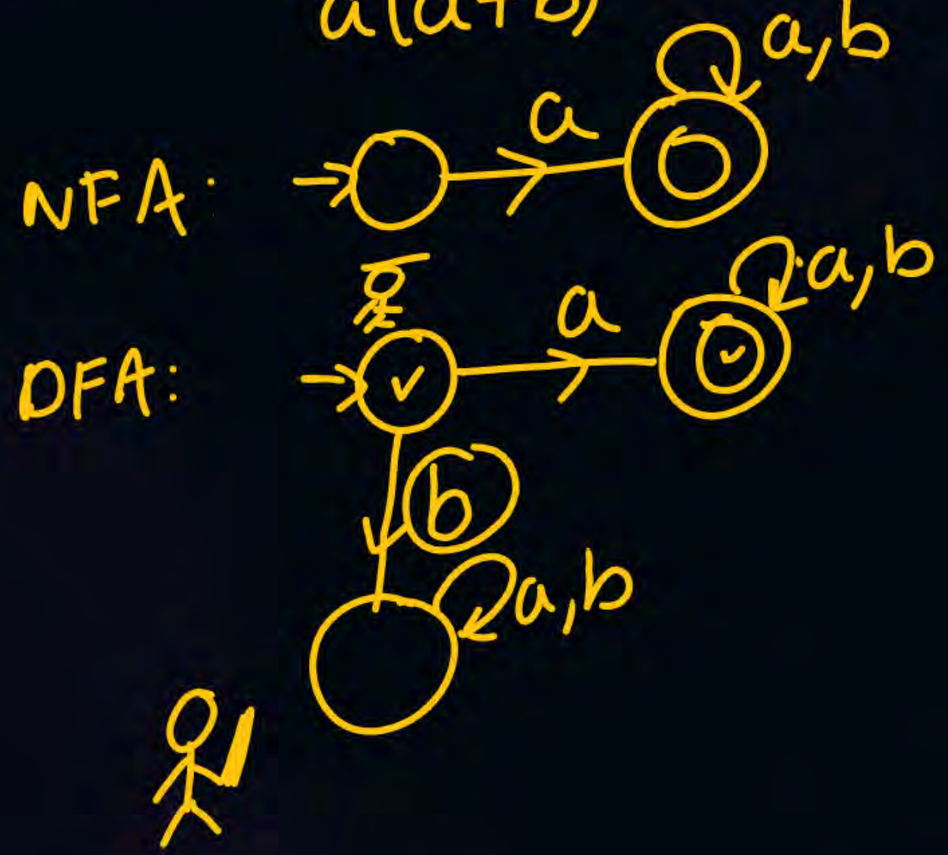
E

$a's \cong 1$ (b) $b's \cong 1$

B, E, H, D, F

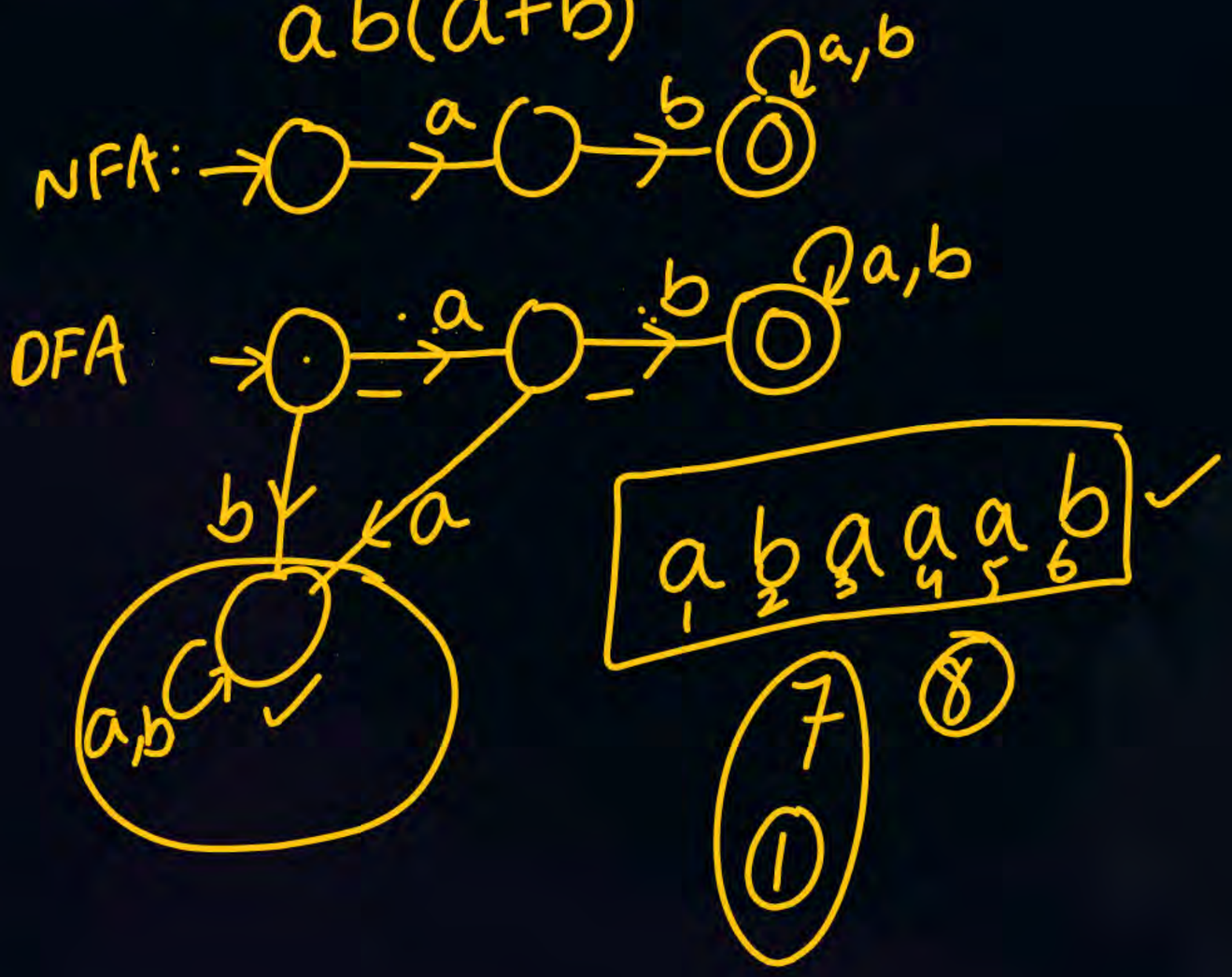
$\Sigma = \{a, b\}$ $L = \{ w / w \text{ starts with } a \}$

$a(a+b)^*$

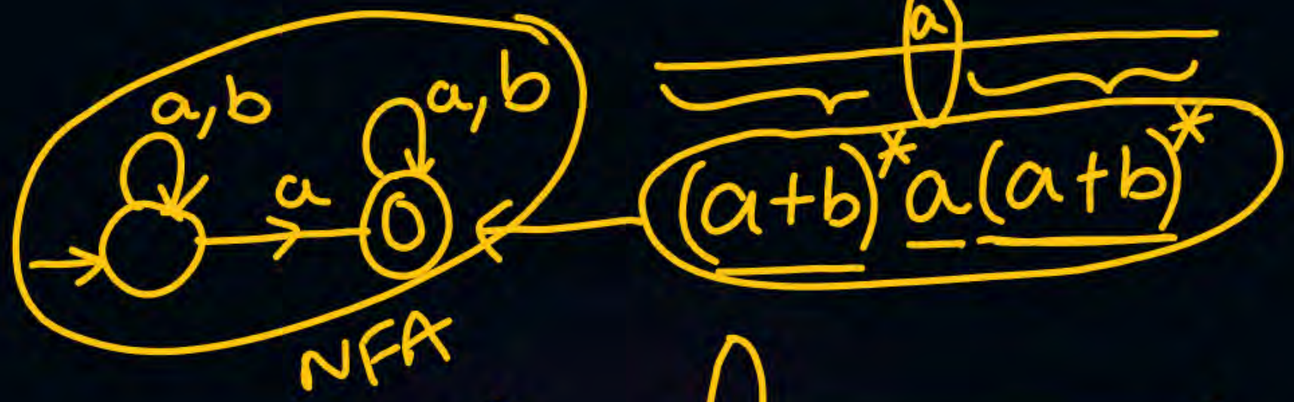


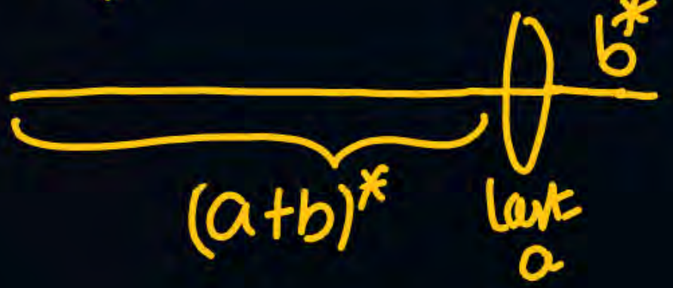
$\Sigma = \{a, b\}$ $L = \{w / w$ starts with $ab\}$

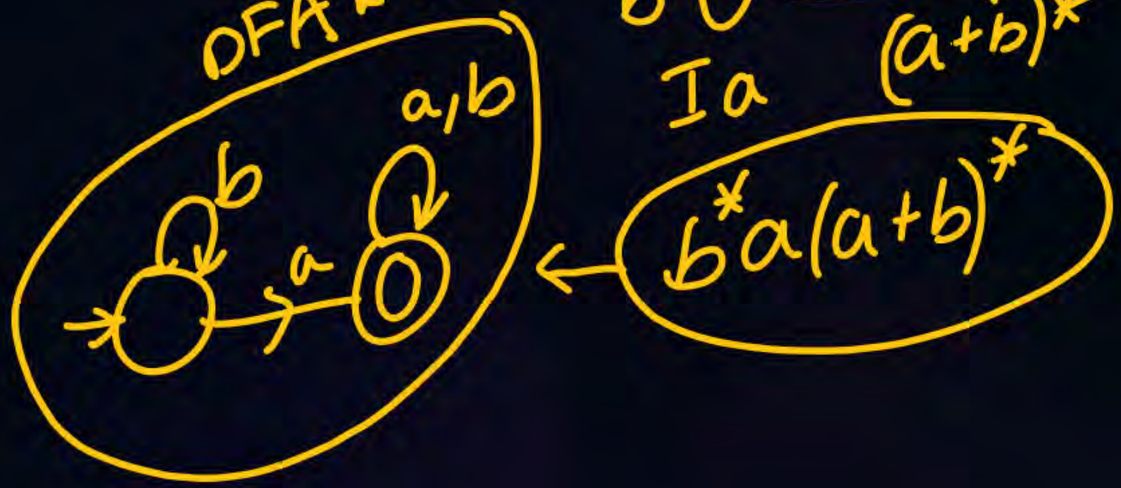
$ab(a+b)^*$

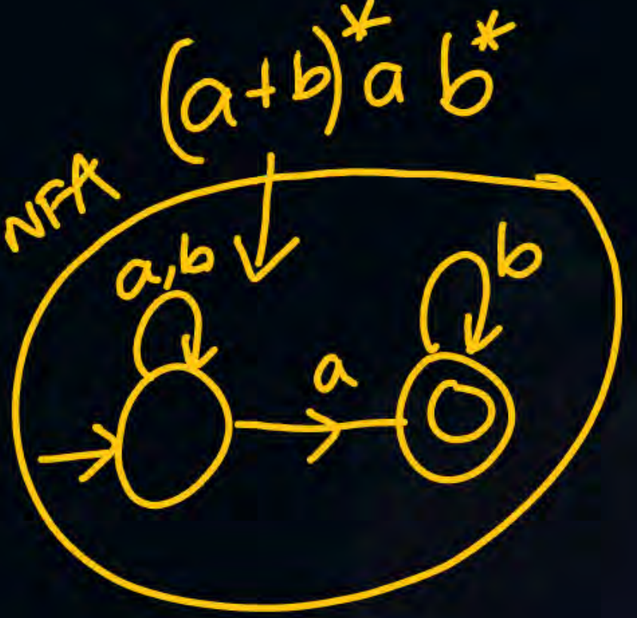


$\Sigma = \{a, b\}$ $L = \{w / w \text{ contains 'a'}\}$









Containing 'a'

$L = \{ a, aa, ab, ba, aaa, aab, \text{-----} \}$



bbbba

THANK - YOU